

Lasso Trading

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1 Draft

Consider a simple asset pricing model in which the equilibrium price of a stochastic asset evolves accordingly to the following law of Motion

$$p_t = p_{t+1}^e + u_t, \quad (1)$$

where p_t is the equilibrium price at time t , p_{t+1}^e is the expected price at time $t+1$ formed at time t before the realization of the equilibrium price and $u_t \sim N(0, \sigma_u^2)$. We assume that agents are boundedly rational and are not aware of the true data generating process. Instead they rely on econometric techniques to form their expectations. Specifically we assume that agents have a linear and stationary perceived law of motion (PLM) for the asset return

$$r_{t+1} = \beta\theta_t + \varepsilon_t, \quad (2)$$

where θ_t is a stochastic signal that agents observe and $\varepsilon_t \sim N(0, \sigma_\varepsilon^2)$.

In order to estimate the parameter agent use Recursive Ordinary Least square of the form

$$\hat{\beta}_t = \frac{\sum_{s=0}^{t-1} r_s \theta_{s-1}}{\sum_{s=0}^{t-1} (\theta_{s-1})^2}, \quad (3)$$

in which the time indexes are consistent with the timeline of events which implies that at time t the last observable return that can be used for estimation is r_{t-1} . The Actual Law of Motion (ALM) for the asset price obtains by substituting the PLM into the equilibrium condition

$$p_t = p_{t+1}^e + u_t = (1 + \hat{\beta}_t \theta_t) p_t^e, \quad (4)$$

where again p_t^e stresses that the current price is not available to agents when making their forecasts. Consistent with their model

$$p_t^e = (1 + \hat{\beta}_t \theta_{t+1}) p_{t-1}, \quad (5)$$

where the use of $\hat{\beta}_t$ is justified by the belief that there is a constant homogenous β and so it makes sense for agents to use the 'smoothed' estimate obtained by considering also the return at time $t-1$ in place of $\hat{\beta}_{t-1}$. This implies an ALM for the current price

$$p_t = (1 + \hat{\beta}_t \theta_t)(1 + \hat{\beta}_t \theta_{t-1}) p_{t-1} + u_t, \quad (6)$$

and it follows that future returns evolve according to

$$r_{t+1} = \hat{\beta}_{t+1}\theta_{t+1} + \hat{\beta}_{t+1}\theta_t + (\hat{\beta}_{t+1})^2\theta_{t+1}\theta_t + \frac{u_t}{p_{t-1}}. \quad (7)$$

1.1 Existence of self-fulfilling equilibria

A self-fulfilling equilibrium requires consistency between the regression coefficient that agents would estimate from the ALM and the actual β implied by their forecast. Formally, let the regression coefficient that would be estimated from the ALM if agents used a constant β be denoted by $\hat{\beta}$, we require

$$\hat{\beta}_t = \beta(\hat{\beta}_t) = \frac{\text{Cov}(r_{t+1}, \theta_t)}{\text{Var}(\theta_t)} \quad (8)$$

where the notation stresses that the population parameter β is a function of the estimated parameter $\hat{\beta}_t$. A self-fulfilling equilibrium is a fixed point of this function, i.e. a solution to the equation $\hat{\beta} = \beta(\hat{\beta})$.

Now

$$\text{Cov}(r_{t+1}, \theta_t) = \hat{\beta} \text{Cov}(\theta_{t+1}, \theta_t) + \hat{\beta} \text{Var}(\theta_t) + \hat{\beta}^2 \text{Cov}(\theta_{t+1}\theta_t, \theta_t) + \text{Cov}\left(\frac{u_t}{p_{t-1}}, \theta_t\right). \quad (9)$$

Given u_t is mean zero and independent of θ , the last term is zero, moreover if we assume that θ_t is iid then every term except the variance is zero and we obtain

$$\hat{\beta} = \hat{\beta}, \quad (10)$$

which implies that every $\hat{\beta}$ leads to a self-fulfilling equilibrium.

1.2 Multiple regressors

We now move to the case in which instead of a single regressor θ_t agents observe a vector of K different signals $\theta_t = (\theta_{1,t}, \theta_{2,t}, \dots, \theta_{K,t})$. To be consistent with dimensions we write the PLM as

$$r_{t+1} = \theta_t \beta + \varepsilon_t, \quad (11)$$

where $\beta = (\beta_1, \beta_2, \dots, \beta_K)^\top$ is a K -dimensional column vector of parameters. The ALM follows accordingly as

$$r_{t+1} = \theta_{t+1}\hat{\beta} + \theta_t\hat{\beta} + (\theta_{t+1}\hat{\beta})(\theta_t\hat{\beta}) + \frac{u_t}{p_{t-1}}. \quad (12)$$

Defining

$$\Theta_t = \begin{pmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_t \end{pmatrix} \quad R_t = \begin{pmatrix} r_2 \\ r_3 \\ \vdots \\ r_{t+1} \end{pmatrix}, \quad (13)$$

as the $t \times K$ matrix of regressors and the t -dimensional vector of returns up to time $t + 1$, the consistency condition requires

$$\hat{\beta} = (\Theta_t^\top \Theta_t)^{-1} \Theta_t^\top R_t = \beta, \quad (14)$$

and the ALM in matrix form is

$$R_t = \Theta_{t+1} \hat{\beta} + \Theta_t \hat{\beta} + (\Theta_{t+1} \hat{\beta})(\Theta_t \hat{\beta}) + \frac{U_t}{P_{t-1}}, \quad (15)$$

where again we use the notation U_t and P_{t-1} to denote the vectors of shocks and prices up to the corresponding time periods.

Now asymptotically we can show that

$$\mathbb{E}[\Theta_t^\top R_t] = \mathbb{E}[\Theta_t^\top \Theta_{t+1}] \hat{\beta} + \mathbb{E}[\Theta_t^\top \Theta_t] \hat{\beta} + \mathbb{E}[\Theta_t^\top (\Theta_{t+1} \hat{\beta})(\Theta_t \hat{\beta})] + \mathbb{E}\left[\Theta_t^\top \frac{U_t}{P_{t-1}}\right] = \mathbb{E}[\Theta_t^\top \Theta_t] \hat{\beta}, \quad (16)$$

which again implies that every $\hat{\beta}$ is a self-fulfilling equilibrium since

$$\mathbb{E}[\hat{\beta}] = \mathbb{E}[(\Theta_t^\top \Theta_t)^{-1}] \mathbb{E}[\Theta_t^\top R_t] = \hat{\beta}. \quad (17)$$

2 A simple example

Consider a reduced form model

$$r_t = \alpha r_t^e + \delta z_{t-1} + \varepsilon_t, \quad (18)$$

in which all quantities are scalars, r_t is the asset return at time t , r_t^e is the expected return formed at time $t - 1$, z_{t-1} is an observable signal at time $t - 1$ and $\varepsilon_t \sim N(0, \sigma_\varepsilon^2)$. The REE requires that $r_t^e = \mathbb{E}_{t-1}[r_t] = \frac{\delta}{1-\alpha}$. We now consider the situation in which agents do not know the exact value of the parameters α and δ and instead form their expectations using a perceived law of motion (PLM) of the form

$$r_t^e = \beta_{t-1} z_{t-1}, \quad (19)$$

where β_{t-1} is the estimate of the regression coefficient obtained using data up to time $t - 1$. Substituting the PLM into the reduced form model we obtain the actual law of motion (ALM)

$$r_t = \alpha \beta_{t-1} z_{t-1} + \delta z_{t-1} + \varepsilon_t. \quad (20)$$

This relationship implies a map between the PLM and ALM captured by the function

$$T(\beta) = \alpha \beta + \delta, \quad (21)$$

and its fixed point coincides with the REE value of the regression coefficient $\beta^* = \frac{\delta}{1-\alpha}$.

We now analyze the stability of this fixed point, which depends on the mechanism used by agents to update their estimates.

2.1 Recursive Least Squares

The first updating mechanism we consider is Recursive Least Squares (RLS), which obtains as a recursive implementation of the standard Ordinary Least Squares (OLS). In OLS, the agent estimates the coefficients of a linear regression model by minimizing the sum of squared residuals. Assuming the general linear model

$$y_i = \beta' x_i + \varepsilon_i, \quad (22)$$

where x_i is the $k \times 1$ vector of regressors, the OLS estimator can be derived from the minimization problem

$$\min_{\beta} \sum_{i=1}^t (y_i - \beta' x_i)^2, \quad (23)$$

with solution

$$\hat{\beta} \equiv b = \left(\sum_{i=1}^t x_i x_i' \right)^{-1} \sum_{i=1}^t x_i y_i. \quad (24)$$

Estimating OLS on the whole available data set, is sometimes referred to as off-line estimation. Another approach is to estimate the model on-line, that is as soon as a new data point arrives. It turns out that the estimator formula allows for a recursive formulation. Define

$$R_t = \frac{1}{t} \left(\sum_{i=1}^t x_i x_i' \right), \quad b_t = \frac{1}{t} R_t^{-1} \sum_{i=1}^t x_i y_i. \quad (25)$$

First we establish a recursion for R_t . Start from

$$R_{t-1} = \frac{1}{t-1} \left(\sum_{i=1}^{t-1} x_i x_i' \right), \quad (26)$$

then

$$\frac{t-1}{t} R_{t-1} = \frac{1}{t} \left(\sum_{i=1}^{t-1} x_i x_i' \right) \rightarrow \frac{t-1}{t} R_{t-1} + \frac{1}{t} x_t x_t' = R_t. \quad (27)$$

That is

$$R_t = R_{t-1} + \frac{1}{t} (x_t x_t' - R_{t-1}). \quad (28)$$

Similarly for b_t we start from

$$b_{t-1} = \frac{1}{t-1} R_{t-1}^{-1} \sum_{i=1}^{t-1} x_i y_i, \quad (29)$$

then notice that

$$(t-1)R_{t-1} = (tR_t - x_t x_t'), \quad (30)$$

which implies that

$$b_{t-1}(tR_t - x_t x_t') = \sum_{i=1}^{t-1} x_i y_i \rightarrow b_{t-1}(1 - t^{-1}R_t^{-1}x_t x_t') = t^{-1}R_t^{-1} \sum_{i=1}^{t-1} x_i y_i, \quad (31)$$

and

$$b_{t-1}(1 - t^{-1}R_t^{-1}x_t x_t') + t^{-1}R_t^{-1}x_t y_t = b_t, \quad (32)$$

which can be rearranged as

$$b_t = b_{t-1} + t^{-1}R_t^{-1}x_t(y_t - x_t' b_{t-1}). \quad (33)$$

The appealing feature of this recursive formulation for our purpose is that it complies with the general formulation of a stochastic recursive algorithm (SRA) of the form

$$\theta_t = \theta_{t-1} + \gamma_t Q(t, \theta_{t-1}, X_t), \quad (34)$$

where θ_t is the parameter vector to be estimated, γ_t is a sequence of gains, $Q(\cdot)$ is a function that depends on the time index, the current parameter estimate and the new data point X_t . Under suitable assumptions about the gain sequence γ_t , and the functional form of $Q(\cdot)$, asymptotic properties are well established. First we show how to recast the RLS updating for our simple model into the SRA form. A small change is required to accomodate for the fact that only lagged values of the parameter estimates are used in $Q(\cdot)$. Define $S_{t-1} = R_t$, then we obtain the updating rules

$$\begin{aligned} \beta_t &= \beta_{t-1} + t^{-1}S_{t-1}^{-1}z_{t-1}[(z_{t-1}(T(\beta_{t-1}) - \beta_{t-1}) + \varepsilon_t)], \\ S_t &= S_{t-1} + t^{-1}\frac{t}{t-1}(z_t^2 - S_{t-1}), \end{aligned} \quad (35)$$

where we used $T(\beta) = \alpha\beta + \delta$.

Now we appropriately define

$$\theta_t = \begin{pmatrix} \beta_t \\ S_t \end{pmatrix}, \quad \gamma_t = t^{-1}, \quad X_t = \begin{pmatrix} z_t \\ z_{t-1} \\ \varepsilon_t \end{pmatrix} \quad (36)$$

Evans and Honapohja (2001) show that the stability of the SRA can be analyzed by studying the small associated ordinary differential equation (ODE)

$$\frac{d\beta}{d\tau} = T(\beta) - \beta, \quad (37)$$

where τ is notational time. This notion, often referred to as E-stability principle, notes that the REE, which is also the fixed point of the ODE, is stable if the ODE is asymptotically

stable. In this case the stability condition is simply given by $\alpha < 1$. In this case RLS learning converges to the REE.

2.2 Weighted Least Squares and Constant Gain Learning

In Weighted Least Squares (WLS), the agent estimates the coefficients of a linear regression model by minimizing a weighted sum of squared residuals. The weights are typically chosen to reflect the relative importance or reliability of different observations. Assuming the linear model (22) the WLS estimator can be derived from the minimization problem

$$\min_{\beta} \sum_{i=1}^t w_{t,i} (y_i - \beta' x_i)^2, \quad (38)$$

with solution

$$\hat{\beta} = \left(\sum_{i=1}^t w_{t,i} x_i x_i' \right)^{-1} \sum_{i=1}^t w_{t,i} x_i y_i. \quad (39)$$

In particular, we are interested in the case where the weights decline geometrically over time, that is

$$w_{t,i} = \gamma(1 - \gamma)^{t-i}, \quad \gamma \in (0, 1). \quad (40)$$

Similar to the OLS case a recursive relationship can be derived.

Define

$$R_t = \gamma \sum_{i=1}^t (1 - \gamma)^{t-i} x_i x_i', \quad b_t = \gamma R_t^{-1} \sum_{i=1}^t (1 - \gamma)^{t-i} x_i y_i \quad (41)$$

First we establish a recursion for R_t . Start from

$$R_{t-1} = \gamma \sum_{i=1}^{t-1} (1 - \gamma)^{t-1-i} x_i x_i', \quad (42)$$

then

$$(1 - \gamma)R_{t-1} = \gamma \sum_{i=1}^{t-1} (1 - \gamma)^{t-i} x_i x_i' \rightarrow (1 - \gamma)R_{t-1} + \gamma x_t x_t' = R_t. \quad (43)$$

That is

$$R_t = R_{t-1} + \gamma (x_t x_t' - R_{t-1}). \quad (44)$$

Similarly for b_t we start from

$$b_{t-1} = \gamma R_{t-1}^{-1} \sum_{i=1}^{t-1} (1 - \gamma)^{t-1-i} x_i y_i, \quad (45)$$

then

$$(1 - \gamma)b_{t-1} = \gamma R_{t-1}^{-1} \sum_{i=1}^{t-1} (1 - \gamma)^{t-i} x_i y_i, \quad (46)$$

and

$$b_{t-1}(R_t - \gamma x_t x_t') = \gamma \sum_{i=1}^{t-1} (1 - \gamma)^{t-i} x_i y_i \rightarrow b_{t-1}(I - \gamma R_t^{-1} x_t x_t') = \gamma R_t^{-1} \sum_{i=1}^{t-1} (1 - \gamma)^{t-i} x_i y_i, \quad (47)$$

since

$$R_{t-1} = \frac{1}{1 - \gamma} (R_t - \gamma x_t x_t').$$

Adding $\gamma R_t^{-1} x_t y_t$ to both sides we obtain

$$b_{t-1}(I - \gamma R_t^{-1} x_t x_t') + \gamma R_t^{-1} x_t y_t = b_t, \quad (48)$$

which can be rearranged as

$$b_t = b_{t-1} + \gamma R_t^{-1} x_t (y_t - x_t' b_{t-1}). \quad (49)$$

In relation to our simple model we get then the same updating rules, with the only difference that the gain is now constant and equal to γ . Given that the gain is not decreasing and that the updating depends on the stochastic term ε_t , the learning rule can not converge to a fixed point. However, under suitable conditions and for small values of γ , and under the same stability condition $\alpha < 1$ it can be shown that the learning dynamics will fluctuate around the REE and in particular that

$$\beta_t \sim \mathcal{N}(\beta^*, \gamma \sigma_\varepsilon^2 (2m_z(1 - \alpha))^{-1}), \quad (50)$$

where $m_z = \mathbb{E}[z_t^2]$, and which makes clear that only in the limit case $\gamma \rightarrow 0$ the variance of the fluctuations vanishes and we recover convergence to the REE.

2.3 Lasso Regression

We now consider the case in which agents use Lasso regression to estimate the parameters of their PLM. The LASSO (Least Absolute Shrinkage and Selection Operator) estimator is defined as the solution to a similar optimization problem

$$\hat{\beta}_{LASSO} = \arg \min_{\beta} \left\{ \frac{1}{2} [(Y - X\beta)'(Y - X\beta)] + \lambda \sum_{j=1}^k |\beta_j| \right\},$$

where $\lambda \geq 0$ is a tuning parameter that controls the amount of regularization. The LASSO estimator can be interpreted as a trade-off between fitting the data well (minimizing the sum of squared residuals) and keeping the model simple (minimizing the absolute values of the coefficients). In general there is no closed form solution for the LASSO estimator, since the absolute value function is not differentiable at zero. However, in the special case where X is an orthonormal matrix, the LASSO estimator can be expressed in closed form.

To do so we can rewrite the optimization problem as

$$\hat{\beta}_{LASSO} = \arg \min_{\beta} \left\{ \frac{1}{2} \left[Y'Y - 2\beta' \underbrace{X'Y}_{\beta'_{OLS}} + \beta' \underbrace{X'X}_I \beta \right] + \lambda \sum_{j=1}^k |\beta_j| \right\},$$

where the underbrackets follow from orthonormality and also notice that the term $Y'Y$ does not depend on β , so it does not change the minimizer. Noticing that $A'A = \sum_{j=1}^k a_j^2$, we can rewrite the problem as

$$\hat{\beta}_{LASSO} = \arg \min_{\beta} \left\{ \sum_{j=1}^k \left(-\beta_j \beta_{OLS,j} + \frac{1}{2} \beta_j^2 + \lambda |\beta_j| \right) \right\}.$$

This problem is separable in the coefficients β_j , so we can solve for each coefficient independently. The optimization problem for each j coefficient is

$$\min_{\beta_j} \left\{ \frac{1}{2} \beta_j^2 - \beta_j \beta_{OLS,j} + \lambda |\beta_j| \right\}.$$

To solve this problem, we can consider two cases:

1. $\beta_{OLS,j} > 0$, then β_j has to be positive, otherwise we could decrease the objective by setting $\beta_j = 0$. In this case, the problem becomes

$$\min_{\beta_j \geq 0} \left\{ \frac{1}{2} \beta_j^2 - \beta_j \beta_{OLS,j} + \lambda \beta_j \right\},$$

and we can now differentiate and set to zero to find the minimum:

$$\frac{\partial}{\partial \beta_j} \left(\frac{1}{2} \beta_j^2 - \beta_j \beta_{OLS,j} + \lambda \beta_j \right) = \beta_j - \beta_{OLS,j} + \lambda = 0 \Rightarrow \beta_j = \beta_{OLS,j} - \lambda.$$

However, we also need to check that this solution is non-negative, so we have

$$\hat{\beta}_{LASSO,j} = \max(0, \beta_{OLS,j} - \lambda).$$

2. $\beta_{OLS,j} < 0$, then β_j has to be negative, otherwise we could decrease the objective by setting $\beta_j = 0$. In this case, the problem becomes

$$\min_{\beta_j < 0} \left\{ \frac{1}{2} \beta_j^2 - \beta_j \beta_{OLS,j} + \lambda (-\beta_j) \right\}.$$

Differentiating and setting to zero, we get

$$\frac{\partial}{\partial \beta_j} \left(\frac{1}{2} \beta_j^2 - \beta_j \beta_{OLS,j} - \lambda \beta_j \right) = \beta_j - \beta_{OLS,j} - \lambda = 0 \Rightarrow \beta_j = \beta_{OLS,j} + \lambda.$$

Again, we need to check that this solution is negative, so we have

$$\hat{\beta}_{LASSO,j} = \min(0, \beta_{OLS,j} + \lambda).$$

Combining both cases, we can write the LASSO estimator for each coefficient as

$$\hat{\beta}_{LASSO} = S(\beta_{OLS}, \lambda) = \text{sign}(\beta_{OLS}) \cdot \max(0, |\beta_{OLS}| - \lambda), \quad (51)$$

where $S(\cdot)$ is the soft-thresholding operator.

In our simple model, we can still use the recursion for the OLS estimator, and then apply the soft-thresholding operator to obtain the LASSO estimate. The mapping between the PLM and ALM now becomes

$$\beta_{OLS} = T(\beta_{LASSO}) = \alpha\beta_{LASSO} + \delta = \alpha S(\beta_{OLS}, \lambda) + \delta, \quad (52)$$

which can be solved case by case:

1. If $\beta_{OLS} > \lambda > 0$, then $\beta_{LASSO} = \beta_{OLS} - \lambda \implies \beta_{OLS} = \beta^* + \frac{\alpha}{1-\alpha}\lambda$.
2. If $\beta_{OLS} < -\lambda < 0$, then $\beta_{LASSO} = \beta_{OLS} + \lambda \implies \beta_{OLS} = \beta^* - \frac{\alpha}{1-\alpha}\lambda$.
3. If $|\beta_{OLS}| \leq \lambda$, then $\beta_{LASSO} = 0 \implies \beta_{OLS} = \delta$.

If we limit ourselves to the case $\delta > 0$, we can summarize the fixed points of the mapping as

$$\beta_{OLS} = \begin{cases} \beta^* + \frac{\alpha}{1-\alpha}\lambda, & \text{if } \delta > \lambda, \\ \delta, & \text{if } \delta \leq \lambda. \end{cases} \quad (53)$$

While this holds in equilibrium, we already discussed that with constant gain learning the estimates will fluctuate around these fixed points. Identically to the previous case, the estimates will be distributed as a normal distribution centered around the fixed point, with variance proportional to the gain parameter γ . So even for $\delta \leq \lambda$, there will be fluctuations in the Lasso estimate around zero, which implies that sometimes the estimates will be exactly zero (when $|\beta_{OLS}| \leq \lambda$) and sometimes they will be non-zero (when $|\beta_{OLS}| > \lambda$). Given the distribution of β_{OLS} , we can compute the probability that the Lasso estimate is exactly zero as

$$\mathbb{P}(|\beta_{OLS}| \leq \lambda) = \Phi\left(\frac{\lambda - \mu_{OLS}}{\sigma_{OLS}}\right) - \Phi\left(\frac{-\lambda - \mu_{OLS}}{\sigma_{OLS}}\right), \quad (54)$$

where μ_{OLS} and σ_{OLS} are the mean and standard deviation of the distribution of β_{OLS} , and $\Phi(\cdot)$ is the cumulative distribution function of the standard normal distribution.

3 The model

We study a standard [Lucas \(1978\)](#) asset pricing model in which we follow the notation in [Adam, Marcet and Nicolini \(2018\)](#). In the baseline we operate under the assumption of risk neutrality and we consider an economy populated by a continuum of agents of mass one who trade a single risky asset. The asset pays a stochastic dividend D_t at each discrete time period $t = 0, 1, 2, \dots$. The dividend and consumption processes are driven by multivariate normally distributed shocks

$$\begin{pmatrix} \log \varepsilon_t^d \\ \log \varepsilon_t^c \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} -s_d^2/2 \\ -s_c^2/2 \end{pmatrix}, \begin{pmatrix} s_d^2 & \rho s_d s_c \\ \rho s_d s_c & s_c^2 \end{pmatrix} \right), \quad (55)$$

The dividend process then follows

$$\frac{D_t}{D_{t-1}} = a \varepsilon_t^d, \quad (56)$$

where $a \geq 1$ is a constant growth factor, which implies that This implies that

$$\mathbb{E}_t [D_{t+1}] = a D_t. \quad (57)$$

Similarly for the aggregate consumption process we postulate that

$$\frac{C_t}{C_{t-1}} = a \varepsilon_t^c, \quad (58)$$

The equilibrium price of the asset is determined by the standard no-arbitrage condition which requires that the price at time t equals the discounted expected value of next period's price plus dividend, that is

$$P_t = \delta \mathbb{E}_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} P_{t+1} + D_{t+1} \right] = \delta \mathbb{E}_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \frac{P_{t+1}}{P_t} \right] P_t + \delta \mathbb{E}_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \frac{D_{t+1}}{D_t} \right] D_t, \quad (59)$$

where δ is the discount factor and $\mathbb{E}_t[\cdot]$ is the expectation operator conditional on information available at time t . Under rational expectations, the equilibrium price can be solved as

$$P_t = \frac{\delta a^{(1-\gamma)} \rho_e}{1 - a^{(1-\gamma)} \rho_e} D_t, \quad (60)$$

where

$$\rho_e = \mathbb{E}_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \frac{D_{t+1}}{D_t} \right] = e^{\gamma(1+\gamma)s_c^2/2} e^{-\gamma\rho s_c s_d}. \quad (61)$$

This implies a constant price-dividend ratio and an ex-dividend gross return which is given by

$$R_{t+1} = \frac{P_{t+1}}{P_t} = \frac{D_{t+1}}{D_t} = a \varepsilon_{t+1}^d. \quad (62)$$

Taking logs we obtain log-returns

$$r_{t+1} = \log(R_{t+1}) = \log(a) + \log(\varepsilon_{t+1}^d) \sim \mathcal{N}(\log(a) - s_d^2/2, s_d^2). \quad (63)$$

Without loss of generality in what follows we set $\log(a) = s_d^2/2$. This implies that under rational expectations log-returns have mean zero and are unpredictable. Instead of assuming that agents know about the true data generating process, we follow the literature in adaptive learning and assume that agents are boundedly rational and form their expectations using econometric techniques. [\[citations here\]](#). Equation (59) shows that agents have to forecast two quantities, the risk adjusted gross return and gross dividend growth. We follow [Adam, Marcet and Nicolini \(2018\)](#) and assume that agents know the process for dividends and only have to learn to forecast gross returns. In this way we can isolate the effects of learning about returns on asset prices. The usual approach to learning is to endow agents with a Perceived Law of Motion (PLM) which is of the same functional form as the Actual Law of Motion (ALM) but with unknown parameters that agents estimate using available data. In this paper we depart from the standard approach and assume that agents use Lasso regression to try and exploit possible log-return predictability. Specifically we assume that agents have a linear and stationary PLM for the asset return

$$r_{t+1} = x_t' \beta + u_t, \quad (64)$$

with $x_t = (x_{1t}, x_{2t}, \dots, x_{kt}) \sim \mathcal{N}(\mathbf{0}, \sigma_x^2 \mathbf{I})$ being k uncorrelated and iid signals available for the agent at time t which they suspect might have predictive power on log-returns. The use of LASSO instead of simple OLS, have some conceptual advantages: [\[Shrinkage, sparsity, PLM goes back to ALM when \$\lambda \rightarrow \infty\$ \]](#)

The PLM is however misspecified, since it implicitly imposes a an ALM which is non-linear. To see this start by noticing that agents PLM about log-returns implies the following PLM of gross returns

$$R_{t+1} = e^{x_t' \beta + u_t}. \quad (65)$$

This implies that gross expected returns follow a log-normal distribution and their conditional forecast if then given by

$$\mathbb{E}_t(R_{t+1}) = \mathbb{E}_t\left(\frac{P_{t+1}}{P_t}\right) = e^{x_t' \beta + \sigma_u^2/2} = \phi e^{x_t' \beta}, \quad (66)$$

by letting $\phi = e^{\sigma_u^2/2}$. We then take the assumption that consumption growth is independent of the return process in the PLM of agents so that

$$\mathbb{E}_t\left[\left(\frac{C_{t+1}}{C_t}\right)^{-\gamma} \frac{P_{t+1}}{P_t}\right] = \mathbb{E}_t\left[\left(\frac{C_{t+1}}{C_t}\right)^{-\gamma}\right] \mathbb{E}_t\left(\frac{P_{t+1}}{P_t}\right) = a^{-\gamma} \phi e^{x_t' \beta}, \quad (67)$$

Plugging this into (59) yields

$$P_t = \frac{\delta a^{(1-\gamma)} \rho_r}{1 - \delta a^{-\gamma} \phi e^{x_t' \beta}} D_t. \quad (68)$$

Now divide by P_{t-1} to obtain actual gross returns given by

$$R_t = \frac{P_t}{P_{t-1}} = \frac{1 - \delta a^{-\gamma} \phi e^{x'_{t-1}\beta}}{1 - \delta a^{-\gamma} \phi e^{x'_t\beta}} \frac{D_t}{D_{t-1}}, \quad (69)$$

and taking logs and moving one step forward we obtain the ALM

$$r_{t+1} = \log(\varepsilon_{t+1}) + \log(1 - \kappa e^{x'_t\beta}) - \log(1 - \kappa e^{x'_{t+1}\beta}). \quad (70)$$

where $\kappa := \delta a^{-\gamma} \phi$.

Clearly this ALM is non-linear, which implies that the linear PLM used by agents is misspecified. In these cases it is unlikely that the learning procedure will converge to the REE and the literature has instead focused on convergence to a Stochastic Consistent Expectation Equilibrium (SCEE) [citations here]. In our case we look for an equilibrium which is cross-moment consistent in the sense that despite their misspecification, there is a statistical self consistency between the empirical relationship they perceive and the one which is implied by the ALM.

In such cross-moment consistent equilibrium the value of β^* is a solution to the equation

$$T(\beta) - \beta = 0, \quad (71)$$

where

$$T(\beta) = COV(x_t, x_t)^{-1} COV(x_t, r_{t+1}), \quad (72)$$

and r_{t+1} is given by (70). Given the assumption about the distribution of x_t we have that $COV(x_t, x_t) = \sigma_x^2 I$. To compute $COV(r_{t+1}, x_t)$ we exploit the fact that x_t is independent of $\log(\varepsilon_{t+1})$ and x_{t+1} , which implies that

$$COV(r_{t+1}, x_t) = COV(\log(1 - \kappa e^{x'_t\beta}), x_t). \quad (73)$$

Two problems arise at this point. The first is that the covariance might not be defined for all values of β since the term inside the logarithm might be negative with positive probability. The second is that there is no closed form solution for the covariance. To deal with this we restrict our analysis to small values of σ_x^2 and such that the term inside the logarithm is positive. Economically speaking this means that predicted log-returns are not too large. [discuss the bound for reasonable values of κ and relate to the risk-free rate] Continuing we therefore use a first-order Taylor expansion of the logarithm around $x'_t = 0$ to obtain

$$\log(1 - \kappa e^{x'_t\beta}) \approx \log(1 - \kappa) + \frac{-\kappa}{1 - \kappa} x'_t\beta, \quad (74)$$

which implies that

$$COV(r_{t+1}, x_t) \approx \frac{-\kappa}{1 - \kappa} COV(x'_t, x_t)\beta, \quad (75)$$

and

$$T(\beta) \approx \frac{-\kappa}{1 - \kappa} \beta. \quad (76)$$

The cross-moment consistent equilibrium is then $\beta^* = 0$.

4 Calibration under Rational Expectations

Under rational expectations the model yields stark predictions that we use to directly match four parameters of the model, namely $[a, s_d, \delta, \gamma]$.

- First, from equation (59) we immediately have that $\mathbb{E}\left(\frac{P_t}{P_{t-1}}\right) = a$, so we set a equal to the average gross return observed in the data.
- Second, we set s_d to match the variance of the gross return, which given the log-normality assumption is given by $Var\left(\frac{P_t}{P_{t-1}}\right) = a^2(e^{s_d^2} - 1)$.
- Third, we set $s_c = \frac{s_d}{7}$ and $\rho = 0.2$ following Adam, Marcet and Nicolini (2018).
- Fourth, we can back out the discount factor δ as an implicit function of the risk aversion parameter γ using the Euler equation for the risk-free rate, which is given by

$$\delta = \mathbb{E}(R_f)^{-1} \left(a^\gamma e^{(-\gamma(1+\gamma)s_c^2/2)} \right). \quad (77)$$

- Finally we retrieve γ by matching the price dividend ratio predicted by the model to the one observed in the data.

$$\mathbb{E}\left(\frac{P_t}{D_t}\right) = \frac{\delta a^{(1-\gamma)} \rho_e}{1 - a^{(1-\gamma)} \rho_e}. \quad (78)$$

where $\rho_e = e^{\gamma(1+\gamma)s_c^2/2} e^{-\gamma\rho s_c s_d}$.

From equation (59) we have that the price-dividend ratio is given by

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